EXPERT SYSTEMS

Expert System Architecture

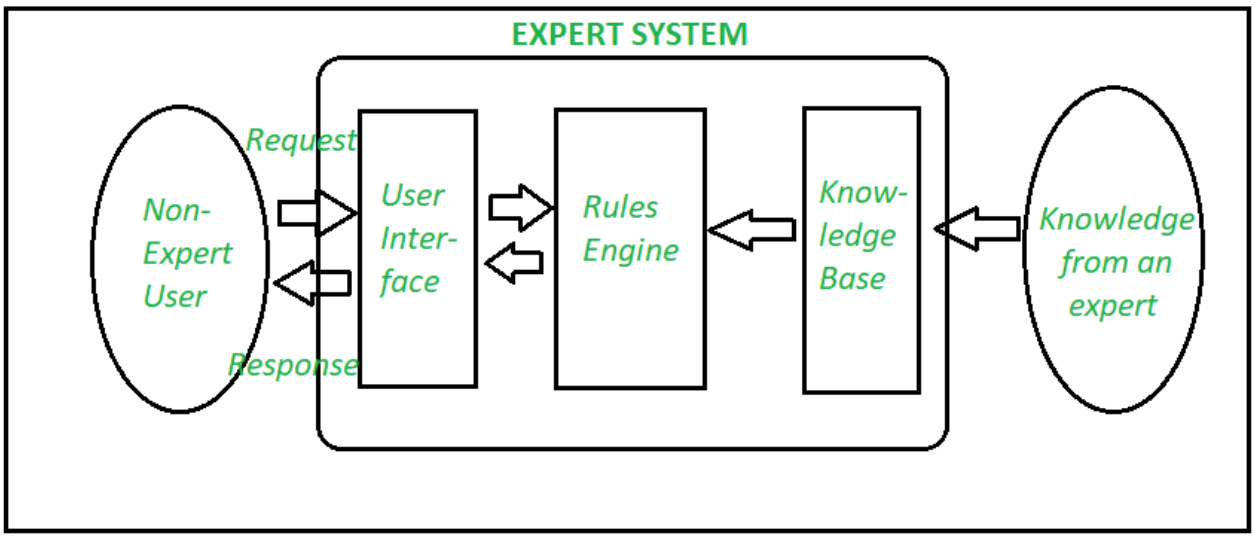
Lecture notes

1. Core Components of an Expert System

A typical expert system has these main components:

1. **Knowledge base:** the domain expertise (facts and rules).
2. **Inference engine:** the reasoning mechanism that applies the knowledge to a problem.
3. **Working memory** (also called the global database or blackboard): the facts about the current case.
4. **User interface:** how the user and the system communicate.
5. **Explanation facility:** justifies how and why the system reached a conclusion.
6. **Knowledge-acquisition facility:** the means of getting expertise into the knowledge base.

People involved: the **domain expert** (source of the knowledge), the **knowledge engineer** (captures and encodes it), and the **end user** (consults the system).



2. Function of Each Component and How They Interact

Component	Function
Knowledge base	Stores domain knowledge, mostly as IF-THEN rules plus facts. It is permanent and reusable.
Inference engine	Selects and fires rules, using forward chaining (data-driven) or backward chaining (goal-driven). It is the "brain" of the system.
Working memory	Holds the facts of the current session: user inputs and conclusions derived so far.
User interface	Takes in questions and data, and presents advice, questions and conclusions in a form the user understands.
Explanation facility	Answers "Why are you asking me this?" and "How did you reach this conclusion?" by tracing the rules that fired.

Component	Function
Knowledge-acquisition facility	Lets experts or knowledge engineers add, edit and validate rules without rewriting the program.

How they interact (one consultation)

1. The user enters a problem or answers questions through the **interface**.
2. These facts are stored in **working memory**.
3. The **inference engine** matches the facts in working memory against the rules in the **knowledge base**.
4. Matching rules fire and produce new facts, which go back into working memory. This cycle (match, select, fire) repeats until a conclusion is reached or no more rules apply.
5. The conclusion is shown to the user through the **interface**.
6. If the user asks "why" or "how", the **explanation facility** reads the rule trace and reports it.
7. Separately, the **knowledge-acquisition facility** updates the knowledge base between sessions.

Key design principle: knowledge is separated from the reasoning mechanism. You can change the rules without touching the inference engine, and reuse an engine (a "shell") across domains.

3. Knowledge Base vs Working Memory

	Knowledge base	Working memory
Contains	General domain knowledge: rules and facts true across cases	Specific facts about the current case
Lifespan	Permanent, persists between sessions	Temporary, cleared or reset after the session
Changed by	Knowledge engineer or expert, through knowledge acquisition	The user's inputs and the inference engine during reasoning
Example	IF fever AND rash THEN suspect measles	Patient has fever = yes, rash = yes
Analogy	A textbook or the expert's training	The notes the expert jots down about this particular patient

4. Explanation and Knowledge-Acquisition Facilities

Explanation facility: purpose

- Builds **user trust**, since people are more likely to accept advice they can follow.
- Lets users **verify** reasoning and spot errors or inappropriate advice.
- Helps **debugging**: the knowledge engineer can see which rules fired and find faulty or missing ones.
- Supports **training**, because novices learn the expert's reasoning.
- Typical queries: **WHY** (why is this information needed?) and **HOW** (how was this conclusion reached?).

Knowledge-acquisition facility: purpose

- Provides the route for **transferring expertise** from human experts (or documents and data) into the system.

- Helps **build, edit and maintain** the knowledge base, which makes it easier to update as the domain changes.
- Often includes consistency and completeness checks (for example, spotting conflicting or redundant rules).
- Addresses the **knowledge-acquisition bottleneck**, the biggest difficulty in building expert systems: experts find it hard to articulate what they know, and capturing it is slow and costly.

Discussion Questions

1. Why is it important to keep the knowledge base separate from the inference engine?
2. Give an example of a fact that belongs in working memory but not in the knowledge base.
3. A doctor refuses to follow a system's diagnosis. Which component helps convince her, and how?